

FINAL EXAM

NAME:Group:

Time to complete the final exam: **90min**. Minimum points to pass the final exam: **18p**.

1. **General concepts.** Which of the two answers is correct/the best choice? Mark your selection by cutting the wrong answer with a line. The first one is done as an example.
 - (a) The acronym for Executable and Linkable Format is *ELF/PE*.
 - (b) Reverse engineering refers to the process of understanding the code's design and specifications starting from the *binary code/high-level language code*. **(2p)**
 - (c) *Avoid side effects/Optimisation* of the source code is a goal of reverse engineering. **(2p)**
 - (d) In an ELF file, pointers to *segments/sections* are used by the linker. **(2p)**
 - (e) The magic number for the PE files is *MZ/PE*. **(2p)**
 - (f) *Static/Dynamic* analysis can understand better what the program is doing while communicating (e.g., sockets, shared memory). **(2p)**
 - (g) *Global Offset Table (GOT)/Position-independent code (PIC)* is a section of a program's memory used to map symbols to their absolute memory addresses. **(2p)**
 - (h) *Hooking/obfuscation* is a technique used to hide the behaviour and/or implementation of an application. **(2p)**
2. **ELF Files.** Look at the image below and answer the questions.

```
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00
  Class:                               ELF64
  Data:                               2's complement, little endian
  Version:                             1 (current)
  OS/ABI:                             UNIX - System V
  ABI Version:                         0
  Type:                                EXEC (Executable file)
  Machine:                             Advanced Micro Devices x86-64
  Version:                             0x1
  Entry point address:                 0x4013e2
  Start of program headers:            64 (bytes into file)
  Start of section headers:           25376 (bytes into file)
  Flags:                               0x0
  Size of this header:                 64 (bytes)
  Size of program headers:            56 (bytes)
  Number of program headers:           9
  Size of section headers:            64 (bytes)
  Number of section headers:          28
  Section header string table index:  27
```

Figure 1: Details of an ELF binary. Source: <https://linux-audit.com>

- (a) Fill in the dots. The ELF file corresponds to a-bit architecture. This is because **(2x2p)**
- (b) Where exactly (in the file) is the header entry for the *.data* section? Briefly explain your computation. **(5p)**

```
[24] .data PROGBITS 00000000006172e0 000172e0 0000000000000100
0000000000000000 **WA** 0 0 8
```

Figure 2: Details on the *.data* section Source: <https://linux-audit.com>

Answer:

- (c) How much memory do all the segments occupy in total? Briefly explain your computation. **(5p) Answer:**

3. Assembly

Given the assembly code below, explain its functionality in 1-2 phrases. Explicitly mention the expected password value for the username ADMIN . **(2x5p)**

```
.LC0:                                mov     rdi, rax
    .string "ADMIN"                call    strcmp
str_un:                             test    eax, eax
    .quad .LC0                    jne     .L2
.LC1:                                mov     eax, 1
    .string "L3Ds21"              jmp     .L3
str_pwd:                            .L2:    mov     eax, 0
    .quad .LC1                    .L3:    leave
fauth(char*, char*):                ret
    push    rbp
    mov     rbp, rsp
    sub     rsp, 16
    mov     QWORD PTR [rbp-8], rdi
    mov     QWORD PTR [rbp-16], rsi
    mov     rax, QWORD PTR str_un[rip]
    mov     rdx, QWORD PTR [rbp-8]
    mov     rsi, rdx
    mov     rdi, rax
    call    strcmp
    test    eax, eax
    jne     .L2
    mov     rax, QWORD PTR str_pwd[rip]
    mov     rdx, QWORD PTR [rbp-16]
    mov     rsi, rdx
main:                                .LC2:    .string "a3bSd1"
    push    rbp
    mov     rbp, rsp
    sub     rsp, 16
    mov     esi, OFFSET FLAT:.LC2
    mov     edi, OFFSET FLAT:.LC0
    call    fauth(char*, char*)
    mov     DWORD PTR [rbp-4], eax
    mov     eax, 0
    leave
    ret
```

Answer:

4. True/False

Respond with true (T) or false (F) and give a short explanation of your choice.

- (a) If the set of permissions for a mapped memory region is "rw-p", then execution is allowed. **(2p) Answer:**
- (b) GOT is a section in the program memory that maps symbols to absolute addresses, while PLT translates function calls to absolute locations. **(2p) Answer:**
- (c) Suppose that an executable ELF file is not compiled as PIC. Then none of the segments (*.text*, *.data*, etc.), the stack, and the dynamic libraries are located at randomised addresses. **(3p) Answer:**

TOTAL available: 40p+5p bonus